

Professional Summary

Researcher in artificial intelligence and data science with extensive experience designing applied models in statistics, machine learning, and deep learning. Holds a B.Sc. in Statistics (Universidad Autónoma Chapingo), an M.Sc. in Mathematical Sciences, and a Ph.D. in Computer Science and Engineering with honors (UNAM, IIMAS), and completed a research stay at the University of Waterloo. Member of Mexico's National System of Researchers (SNII). Over the past five years, work has concentrated on explainable generative AI for medical imaging, environmental health, and language models, as well as inverse problems and scientific computing. Currently a research professor at INFOTEC, teaching graduate courses and supervising doctoral theses in data science. Research output includes articles in international journals and conferences and numerous distinctions, among them the Best Paper Award at MICA 2023, first place in the IJCNN 2025 "Learning with Noisy Graph Labels" competition, third place in the 2024 AFIRME-FUNAM Research Award, and a Top 10 finish in Intel's 2025 "Accelerating Mexico with Artificial Intelligence" program. Experience also includes consulting in risk modeling and analysis, as well as projects for government institutions and productive sectors, applying AI and data science to public health, the environment, energy, and finance.

Education

- 2015-2020 **Ph.D. in Computer Science and Engineering**, UNAM, IIMAS, Honors
Scientific computing, optimization, machine learning, and data science.
- 2017-2018 **Research stay (visiting student)**, University of Waterloo, Canada
Scientific computing, optimization, and hydraulic tomography.
- 2011-2013 **M.Sc. in Mathematical Sciences**, UNAM, IIMAS
Scientific computing, statistics, and probability.
- 2003-2007 **B.Sc. in Statistics**, Universidad Autónoma Chapingo

Professional Experience

- 2024-present **Tenured Research Professor**, INFOTEC
Research in data science and artificial intelligence applied to health; teaching and thesis supervision in the Master's and Ph.D. programs in Data Science.
- 2022-2024 **Postdoctoral Researcher**, CECAv-UNAM
Explainable generative AI for medical image classification and analysis (origin of PumaMedNet-CXR).
- 2021-2022 **Associate Researcher**, UPIITA-IPN
Predictive risk modeling and disease traceability using data from the Ministry of Health.
- 2020-2021 **Associate Researcher**, AI Consortium (CONACyT)
Hybrid statistical-ML model to predict air pollutant concentrations in Mexico City.
- 2016-2020 **Machine Learning Developer**, Fintech Proyecto-PyME
Loan default prediction for small and medium-sized enterprises.
- 2008-2020 **Statistical Consultant and Data Scientist**, Minnova Consulting / Data FusionLab
Risk modeling and analysis.
- Earlier Hydraulic tomography (University of Waterloo, 2017-2018); characterization of naturally fractured reservoirs (PEMEX / CONACyT-SENER, 2013-2014); reconstruction and sampling of the 1930 Census (INEGI, 2012-2015); national survey on climate change (INECC-PNUD, 2015-2016).

Teaching and Thesis Supervision

- Teaching Time Series Analysis and Statistics (Master's in Data Science, INFOTEC, 2024-present); Research Seminar and Thesis Seminar (Ph.D. in Data Science, INFOTEC, 2025); Web Analytics (Master's in Data Science and Information, INFOTEC, 2022-2023); Time Series (B.Sc. in Data Science, UNAM, 2022-2023).
- Curriculum design Technical lead for the syllabi of Statistics, Biological Network Analysis, and Time Series Analysis for INFOTEC's graduate program in Data Science and Information.
- Thesis supervision Ph.D. in Data Science and Information, INFOTEC:
 - Explainable and robust machine learning: causal inference for critical applications (advisor).

- Knowledge extraction, natural language processing, and language models (LLMs) (advisor).
- Application of multimodal models in portfolio optimization and financial analysis (advisor).
- Forecasting for time series analysis using machine learning (advisor).
- Ad-hoc statistical methodology for continued process verification (co-advisor).

Awards and Distinctions

- 2026 **Level 1, National System of Researchers (SNII)**, *SECIHTI*, In effect from 2027
- 2026 **1st Place, MSLG-SPA Task (bidirectional Mexican Sign Language-Spanish translation)**, *International Competition, IberLEF 2026*
- 2026 **2nd Place, Rest-Mex 2026 Task**, *International Competition, IberLEF 2026*
- 2025 **Top 10, “Accelerating Mexico with Artificial Intelligence” Program**, *Intel (national competition)*
- 2025 **1st Place, “Learning with Noisy Graph Labels” International Competition**, *IJCNN 2025 (INNS)*, Rome, Italy
- 2025 **2nd Place, Rest-Mex 2025 Task**, *International Competition, IberLEF 2025 (CIMAT / SEPLN)*
- 2024 **3rd Place, Research Category, AFIRME-FUNAM Award**, *UNAM, AFIRME, and Fundación UNAM*
- 2024 **Fellow, AI Talent Development Program 1000**, *INFOTEC-Huawei*, Hangzhou, China
- 2023 **1st Place, Best Paper Award, 22nd MICAI**, *Mexican Society for Artificial Intelligence*
- 2023 **Member of SNII, Candidate**, *SECIHTI*
- 2021 **Silver Medal (Top 3%), Google Smartphone Decimeter Challenge**, *Kaggle*
- 2017 **Fellow, Emerging Leaders in the Americas Program**, *Canada*
- 2016 **2nd Place, Best Master’s Thesis in Statistics**, *“Francisco Aranda-Ordaz” National Award (AME)*

Publications

- Muraharirao, P.C., Kambhammettu, B., Minutti-Martinez, C., Illman, W.A., et al. **“PI-XNET: Physics-informed Multi-task Neural Networks for Joint Mapping of Hydraulic Conductivity and Preferential Flow Paths in Fractured Aquifers”**. *Advances in Water Resources*, 105470 (2026).
- Morales, M.P., Olveres, J., Escalante-Ramírez, B., Díaz, R.R., Minutti-Martinez, C. **“Stress-testing Medical Vision Models with Images Edited by Controllable Inversion Algorithms”**. *Optics, Photonics, and Digital Technologies for Imaging Applications IX*, SPIE (2026).
- Minutti-Martinez, C., Escalante-Ramírez, B., Olveres, J. **“Cyclo-VGAE: Dual-Mechanism Approach to GNN Robustness Against Noisy Labels”**. *International Joint Conference on Neural Networks (IJCNN)*, IEEE, pp. 1-8 (2025).
- Lorenzo-Landa, G., Minutti-Martinez, C. **“Forecasting an Emerging Market Stock Index: a Comparative Study of Classical and Deep Learning Models”**. *CONISOFT*, IEEE (2025).
- Minutti-Martinez, C., Mata-Rivera, M.F., Arellano-Vazquez, M., Escalante-Ramírez, B., Olveres, J. **“Air Pollution, Socioeconomic Status, and Avoidable Hospitalizations: A Multifaceted Analysis”**. *Mathematical and Computational Applications*, 30(4), 69 (2025).
- Arellano-Vázquez, M., Zamora-Machado, M., Robles-Pérez, M., Minutti-Martinez, C., Jaramillo, O. **“Dynamic Wind Condition Detection in Baja California, Mexico: A Machine Learning Approach for Improved Wind Management”**. *IEEE Latin America Transactions*, 23(5), 387-396 (2025).
- Minutti-Martinez, C., Mata-Rivera, M.F., Arellano-Vazquez, M., Escalante-Ramírez, B., Olveres, J. **“Air Pollution, Socioeconomic Status, and Avoidable Hospitalizations in Mexico City: A Multifaceted Analysis”**. *Lecture Notes in Computer Science (MICAI)*, Springer, pp. 73-86 (2025).
- Minutti-Martinez, C., Escalante-Ramírez, B., Olveres, J. **“Multitask Analysis of Spanish Travel Reviews: Sentiment, Destination, and Topic Classification with RoBERTa and LLaMA Ensembles”**. *CEUR Workshop Proceedings, IberLEF-SEPLN* (2025).
- Minutti-Martinez, C. **“Unraveling the Complex Interplay Between Socioeconomic Status, Air Pollution, and Heart Disease Hospitalizations in an Urban Population”**. *21st International Conference on Electrical Engineering, Computing Science and Automatic Control (CCE)*, IEEE (2024).
- Minutti-Martinez, C. **“Aproximación práctica a los métodos de selección de portafolios de inversión”**. *arXiv:2410.11070* (2024).

- Carrillo, R., Minutti, C., Lagunes, P. **“A Comprehensive Methodology for Performing Continued Process Verification”**. IEEE 37th International Symposium on Computer-Based Medical Systems (CBMS) (2024).
- Minutti-Martinez, C., Escalante-Ramírez, B., Olveres, J. **“Enhancing Interpretability and Bias Control in Deep Learning Models for Medical Image Analysis Using Generative AI”**. Proceedings of SPIE (2024).
- Minutti-Martinez, C., Escalante-Ramírez, B., Olveres-Montiel, J. **“PumaMedNet-CXR: An Explainable Generative Artificial Intelligence for the Analysis and Classification of Chest X-Ray Images”**. Computación y Sistemas, 27(4), 909-920 (2023).
- Minutti-Martinez, C., Galindo, A., Valdez-Garduño, L.F., Mata-Rivera, M.F. **“Exploring Nonlinear Effects of Air Pollution on Hospital Admissions by Disease Using Gradient Boosting Machines”**. 19th CCE, IEEE (2022).
- Minutti-Martinez, C., Arellano-Vázquez, M., Zamora-Machado, M. **“A Hybrid Model for the Prediction of Air Pollutants Concentration, Based on Statistical and Machine Learning Techniques”**. Lecture Notes in Computer Science (MICAI), Springer (2021).
- Minutti, C., Illman, W.A., Gomez, S. **“A New Inverse Modeling Approach for Hydraulic Conductivity Estimation Based on Gaussian Mixtures”**. Water Resources Research, 56 (2020).
- Arellano-Vázquez, M., Minutti-Martinez, C., Zamora-Machado, M. **“Automated Characterization and Prediction of Wind Conditions Using Gaussian Mixtures”**. Lecture Notes in Computer Science (MICAI), Springer (2020).
- Minutti, C., Illman, W.A., Gomez, S. **“An Algorithm for Hydraulic Tomography Based on a Mixture Model”**. Lecture Notes in Computer Science (ICCS), Springer, pp. 471-486 (2019).
- Minutti, C., Gomez, S., Ramos, G. **“A Machine-Learning Approach for Noise Reduction in Parameter Estimation Inverse Problems, Applied to Characterization of Oil Reservoirs”**. Journal of Physics: Conference Series, 1047(1), 012010 (2018).
- Ramos, G., Carrera, J., Gómez, S., Minutti, C., Camacho, R. **“A Stable Computation of Log-Derivatives from Noisy Drawdown Data”**. Water Resources Research, 53(9), 7904-7916 (2017).
- Minutti, C., Ramos, G., Gomez, S., Camacho, R., Vázquez, M., Castillo, N. **“Robust Characterization of Naturally Fractured Carbonate Reservoirs Through Sensitivity Analysis and Noise Propagation Reduction”**. SPE Latin America and Caribbean Heavy and Extra Heavy Oil Conference (2016).
- Camacho-Velázquez, R., Gómez, S., Vásquez-Cruz, M.A., Fuenleal, N.A., Castillo, T., Ramos, G., Minutti, C., et al. **“Well-Testing Characterization of Heavy-Oil Naturally Fractured Vuggy Reservoirs”**. SPE Heavy and Extra Heavy Oil Conference: Latin America (2014).
- Camacho-Velázquez, R., Fuenleal, N., Castillo, T., Gómez, S., Ramos, G., Minutti-Martínez, C., et al. **“Avances en la caracterización integral de un yacimiento naturalmente fracturado vugular, el caso Ayatsil-Tekel”**. Ingeniería Petrolera, 54(10), 583-601 (2014).
- Zamudio, F.J., Arana, R.I., Jiménez, J., Minutti, C., Santibáñez, J., McCaa, R. **“Muestreo Probabilístico para la Recuperación de los Microdatos del Censo General de Población de 1930”**. Instituto Nacional de Estadística y Geografía, INEGI (2015).

International Conferences

- **“Cyclo-VGAE: Dual-Mechanism Approach to GNN Robustness Against Noisy Labels”**. IJCNN 2025, Rome, Italy.
- **“Unraveling the Complex Interplay Between Socioeconomic Status, Air Pollution, and Heart Disease Hospitalizations”**. 21st CCE 2024, Mexico City.
- **“Air Pollution, Socioeconomic Status, and Avoidable Hospitalizations in Mexico City”**. 6th CIAPP 2024, INAOE, Puebla.
- **“Enhancing Interpretability and Bias Control in Deep Learning Models for Medical Image Analysis Using Generative AI”**. SPIE Photonics Europe 2024, Strasbourg, France.
- **“PumaMedNet-CXR: An Explainable Denoising Autoencoder for the Analysis and Classification of Chest X-Ray Images”**. 22nd MICAI 2023, Mérida, Mexico.
- **“Exploring Nonlinear Effects of Air Pollution on Hospital Admissions by Disease Using Gradient Boosting Machines”**. 19th CCE 2022, Mexico City.
- **“Automated Prediction of Air Pollutant Concentrations in Mexico City, Using Artificial Intelligence Methods”**. 20th MICAI 2021, Mexico City.
- **“An Algorithm to Perform Hydraulic Tomography Based on a Mixture Model”**. ICCS 2019, Faro, Portugal.

- “An Approach for Noise Reduction in Inverse Problems, Applied to Well-Test Data”. 9th Inverse Problems in Engineering 2017, Waterloo, Canada.
- “Improved Characterization of Naturally Fractured-Vuggy Carbonate Reservoirs Through a Statistical Multivariate Analysis” (poster). Control and Inverse Problems in PDE 2016.
- “A Methodology for the Characterization of Naturally Fractured-Vuggy Carbonate Reservoirs Using Statistical Methods”. 8th Inverse Problems: Modeling and Simulation 2016, Fethiye, Turkey.

National Conferences, Outreach, and Media Engagements (Selection)

- “From Image to Report Without Hallucinations: Multimodal Explainability for Chest X-Ray Analysis”. 22nd Mexican Symposium on Medical Image Computing, INPer, Mexico City (2026).
- “Multimodal AI Platform for Explainable Analysis and Diagnosis Using Medical Images”. INFOTEC (2026).
- “Medical Image Classification with Generative AI and Dynamic Interpretability”. UAM Iztapalapa (2026).
- “Interpretability in Medical Image Analysis with Deep Learning and Generative AI” (keynote lecture). CIMAT (2025).
- “PumaMedNet Advances: A Generative Architecture with Dynamic Interpretability for Medical Image Classification”. MEXCAS, UNAM (2025).
- “Artificial Intelligence and Data Science in the Service of Health Through Explainable Models”. COPARMEX (2025).
- “Congress on Technological Innovation and Digital Transformation” (panel discussion). ATDT (2025).
- “The Big Data Era: Data That Knows You Better Than You Know Yourself” (radio and podcast). Instituto Mexicano de la Radio (IMER) (2025).
- “An Explainable Generative AI for the Analysis and Classification of Chest X-Ray”. INFOTEC (2024).
- “Responsible and Transparent Data Science and AI in Health”. Universidad Iberoamericana, Mexico City (2024).
- “Responsible and Transparent Data Science and AI in Health; AI and Big Data: Medical Applications”. INFOTEC (2024).
- “Responsible and Transparent Data Science and AI in Health” (Data and AI Analysis Seminar). UAM Iztapalapa (2024).
- “Interview on Ingeniería en Marcha About the MICA Best Paper Award” (radio). Radio UNAM (2024).
- “Enhancing Interpretability and Bias Control in Deep Learning Models for Medical Image Analysis Using Generative AI” (poster). First Congress on Digital Health, UNAM School of Medicine (2024).
- “1st UNAM Symposium on Generative Artificial Intelligence in Education” (panelist). UNAM (2023).
- “Building Socioeconomic and Air Pollution Indicators and Their Relationship with Health Effects” (Virtual Colloquium on Data Science). UNAM (2023).
- “Building Socioeconomic and Air Pollution Indicators and Their Relationship with Health Problems”. UPIITA-IPN (2022).
- “A Hybrid Model for Predicting Air Pollutants Based on Statistical and Machine Learning Techniques”. UNAM (2022).
- “Can We Trust the Numbers? Misleading Statistics” (science café). CIMAT (2022).
- “Hydraulic Tomography Through a Gaussian Mixture Model”. 7th Metropolitan Congress on Numerical Modeling and Simulation (2019).
- “Improved Characterization of Oil Reservoirs Using a Fractal Model and Statistical Methods”. 5th Metropolitan Congress on Numerical Modeling and Simulation (2017).
- “Developing a Methodology for the Characterization of Oil Reservoirs Using Statistical and Numerical Methods”. 31st International Statistics Forum (2016).

Intellectual Property and Technology Transfer

Software registration “Web-based system for modeling mortality risk and hospital severity associated with medical conditions” (granted, 2023; with M. F. Mata-Rivera).

Peer Review and Academic Evaluation

Reviewer	IEEE Transactions on Emerging Topics in Computational Intelligence (2026); “Bioinformatics and Computational Biology” track, ROPEC (2024 and 2025).
Project evaluator	Call for Academic Events to Strengthen International Scientific and Humanistic Collaboration, SECIHTI (2025).

Courses and Certifications

Huawei Certified AI Engineer (HCIA-AI) and Advanced AI Training Certificate (Huawei, 2024); Machine Learning (Stanford University, 2015); Academic Writing (UNAM, 2015); Summer School on Parallel Supercomputing and GPUs (UNAM, 2014); Data or Specimens Only Research and Conflicts of Interest (CITI Program, 2023).

Languages

Spanish	Native
English	Advanced (certified)

Software and Technical Skills

Languages	C/C++, Fortran+MPI, Python, Bash Scripting.	Web	HTML, PHP, JavaScript, XML, AWS.
Scientific Computing	R, Python (NumPy, SciPy, Pandas, PyTorch), SAS, Octave, Matlab, SciLab, Maxima.	Databases	MySQL, SQLite.
Publishing	LaTeX, Scribus.	Graphics	GIMP, Blender, Inkscape.
Research areas	Artificial intelligence, data science, explainable AI (XAI), generative AI, health, computational intelligence, statistical modeling, and scientific computing.		